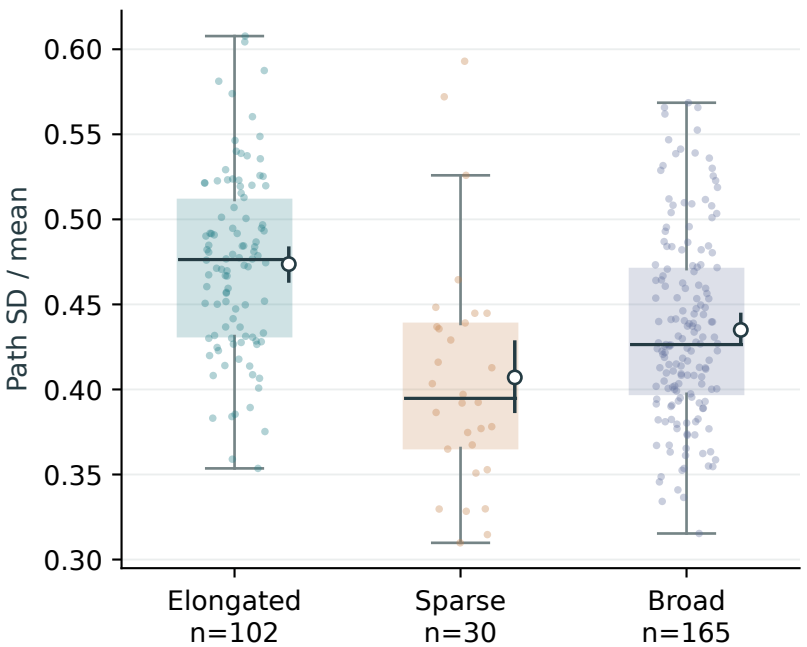


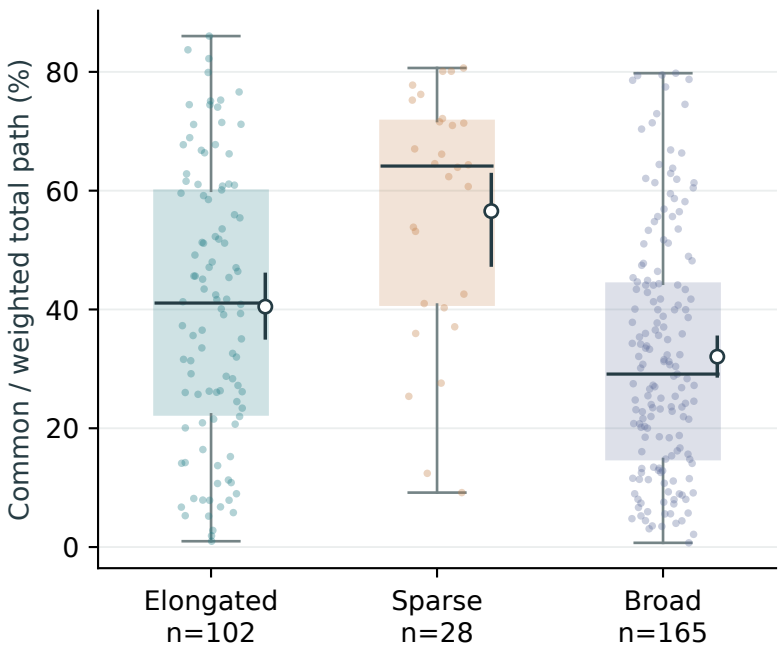
River form → structural mechanisms → DOC evidence

Three original forms retained; continuous structure resolves within-form differences

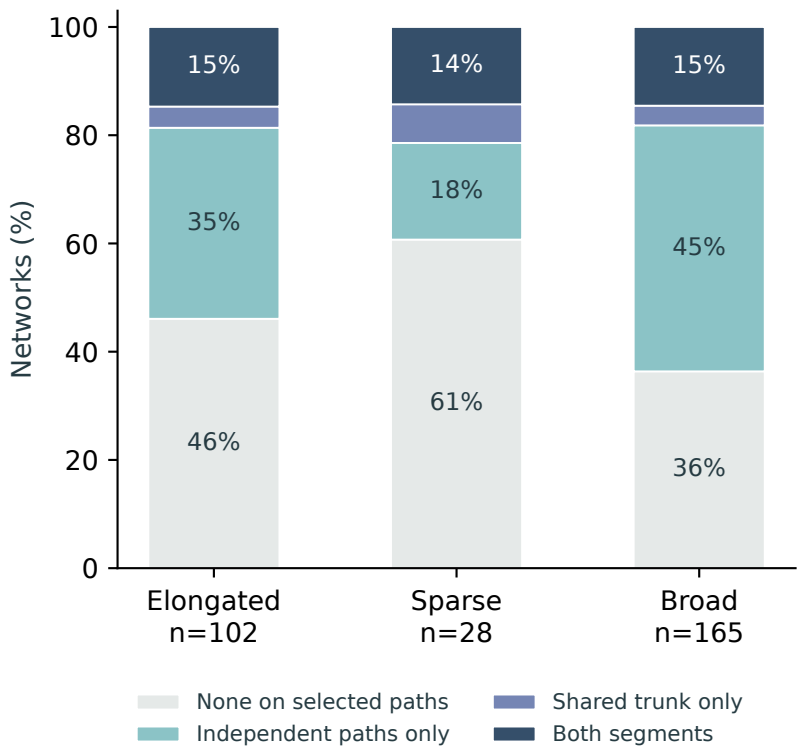
a Whole-network arrival dispersion



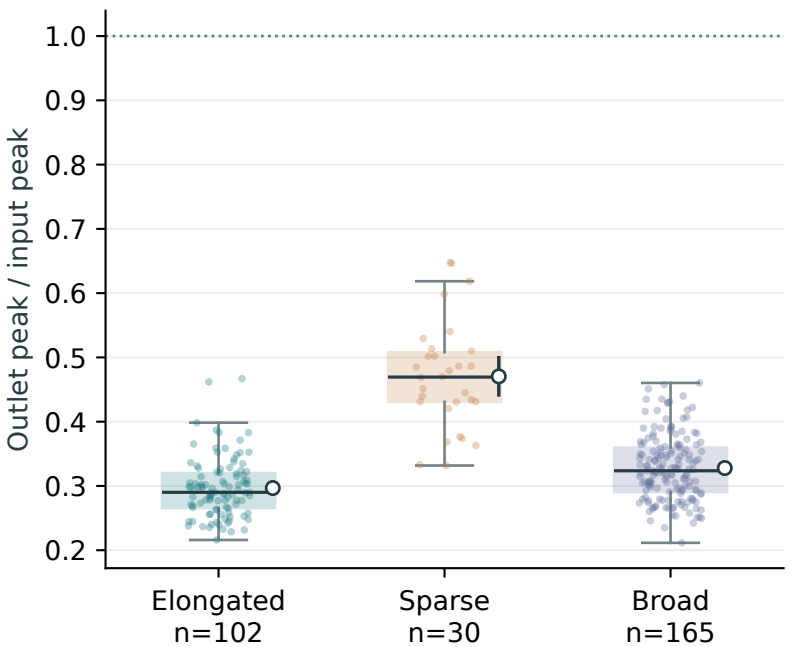
b Shared path after the major confluence



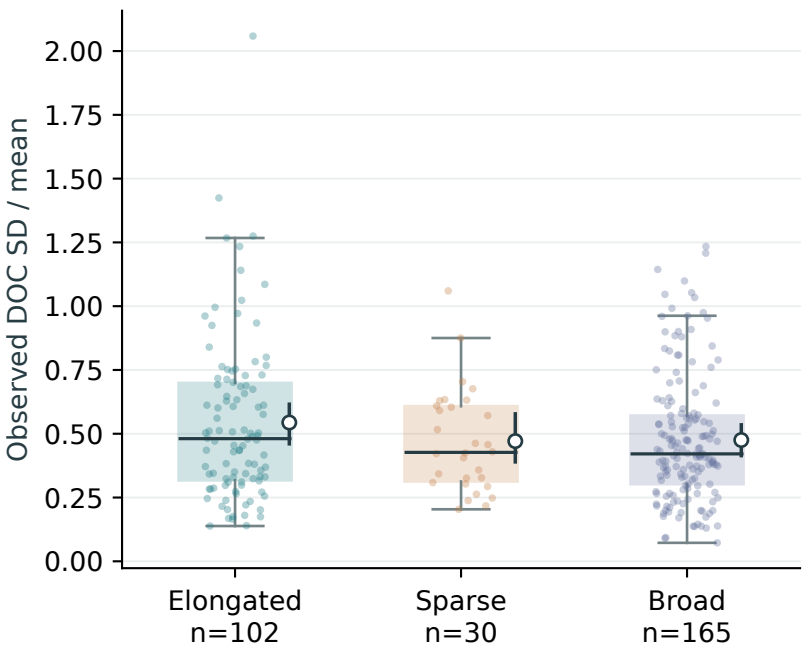
c Where is mapped storage?



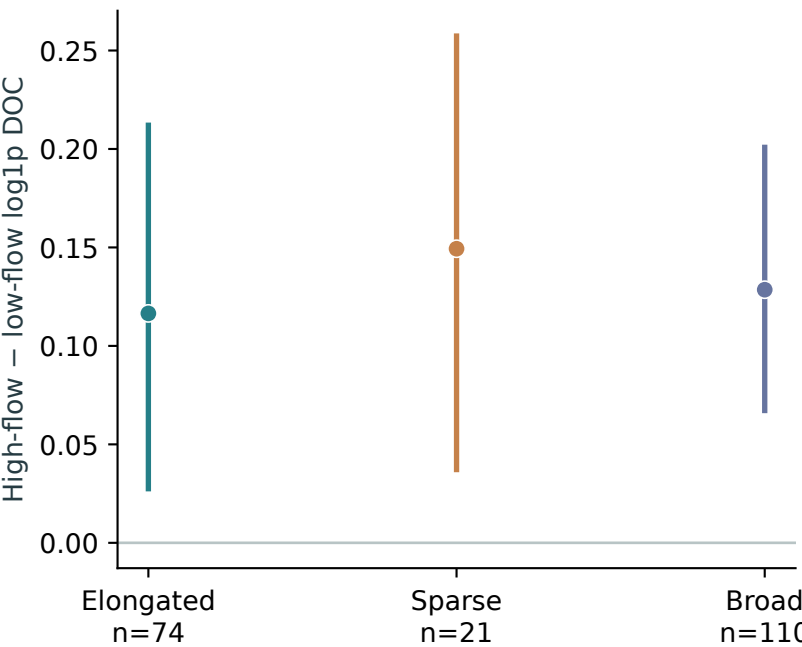
d Identical-input structural response



e Measured monthly DOC variation



f Season/year-adjusted flow response



a/b/d/e: dots are station instances; boxes show quartiles; open points/lines are means and HUC4 95% intervals. c: selected representative paths only. d uses controlled reach-midpoint forcing (mean arrival = 1); e/f use monthly observations, not event peaks. Intervals use 5,000 regional resamples.